

Four-Dimensional Parameter Expansion Study

Generated by opt_4dim.py

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Purpose

This report extends the 3D side study from (k, δ, x) to (k, δ, x, y) , testing all selected pairs (x, y) from the remaining Layer 2 parameter set

$$\{q, \alpha, \beta, L, a, z_{\text{trend}}, \rho\}.$$

Pairs involving ρ use online smoothing inside each fold, because ρ does not affect the static single-fold strategy directly.

Dataset and Protocol

Data source	Yahoo adjusted SPY
Symbol	SPY
Lookback	5 years
Bar interval	1d
Bars	1254
Train size	252 bars
Test size	63 bars
Cost per side	0.0 bps
Objective	$R^m + D^m + 0.15G$

Search Grid

This run used `--fast`: Yes: reduced screening grid for tractability. The 4D grid can grow quickly because each experiment searches (k, δ, x, y) , so the fast run should be read as a sensitivity screen rather than a final exhaustive optimization.

Parameter	Values searched
k	0.8, 1.2
delta	1, 3
q	0.0, 5e-08
alpha	0.03, 0.06
beta	0.8, 0.88
L	30, 60
a	0.02, 0.04
z_trend	0.0, 0.25
rho	0.1, 0.25

Stitched Walk-Forward Results

Experiment	Optimized	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	Final selected
baseline	k, delta	69.25%	77.29%	-4.54%	-18.76%	0.90	0.80	5	33.20	0.00%	k=1.2, delta=3
q+alpha	k, delta, q, alpha	63.22%	77.29%	-7.93%	-18.76%	0.85	0.74	5	32.20	-6.02%	k=0.8, delta=1, q=0, alpha=0.03
q+beta	k, delta, q, beta	61.34%	77.29%	-9.00%	-18.76%	0.83	0.73	5	32.20	-7.91%	k=0.8, delta=1, q=0, beta=0.8
q+L	k, delta, q, L	60.38%	77.29%	-9.54%	-18.76%	0.83	0.72	6	32.17	-8.87%	k=0.8, delta=1, q=5.0e-08, L=30
q+a	k, delta, q, a	67.73%	77.29%	-5.39%	-18.76%	0.93	0.79	10	27.40	-1.52%	k=1.2, delta=1, q=5.0e-08, a=0.02
q+z_trend	k, delta, q, z_trend	63.22%	77.29%	-7.93%	-18.76%	0.85	0.74	5	32.20	-6.02%	k=1.2, delta=3, q=0, z_trend=0

q+rho	k_init, delta_init, q, rho	70.07%	77.29%	-4.07%	-18.76%	0.95	0.81	9	26.33	0.82%	k_init=0.8, delta_init=3, q=5.0e-08, rho=0.1
alpha+beta	k, delta, alpha, beta	65.62%	77.29%	-6.58%	-18.76%	0.87	0.77	5	32.20	-3.63%	k=0.8, delta=3, alpha=0.03, beta=0.8
alpha+L	k, delta, alpha, L	64.63%	77.29%	-7.14%	-18.76%	0.87	0.76	6	32.17	-4.61%	k=0.8, delta=1, alpha=0.03, L=30
alpha+a	k, delta, alpha, a	59.17%	77.29%	-10.22%	-18.76%	0.82	0.70	9	29.44	-10.07%	k=1.2, delta=3, alpha=0.03, a=0.02
alpha+z_trend	k, delta, alpha, z_trend	66.64%	77.29%	-6.00%	-18.76%	0.88	0.78	6	27.17	-2.60%	k=1.2, delta=3, alpha=0.06, z_trend=0
alpha+rho	k_init, delta_init, alpha, rho	70.07%	77.29%	-4.07%	-18.76%	0.95	0.81	9	26.33	0.82%	k_init=0.8, delta_init=3, alpha=0.06, rho=0.1
beta+L	k, delta, beta, L	64.54%	77.29%	-7.19%	-18.76%	0.87	0.76	6	32.17	-4.71%	k=0.8, delta=1, beta=0.8, L=30
beta+a	k, delta, beta, a	58.77%	77.29%	-10.45%	-18.76%	0.82	0.70	9	29.44	-10.48%	k=1.2, delta=3, beta=0.8, a=0.02
beta+z_trend	k, delta, beta, z_trend	66.64%	77.29%	-6.00%	-18.76%	0.88	0.78	6	27.17	-2.60%	k=1.2, delta=3, beta=0.8, z_trend=0
beta+rho	k_init, delta_init, beta, rho	70.07%	77.29%	-4.07%	-18.76%	0.95	0.81	9	26.33	0.82%	k_init=0.8, delta_init=3, beta=0.8, rho=0.1
L+a	k, delta, L, a	59.17%	77.29%	-10.22%	-18.76%	0.82	0.70	9	29.44	-10.07%	k=1.2, delta=1, L=60, a=0.02
L+z_trend	k, delta, L, z_trend	69.32%	77.29%	-4.50%	-18.76%	0.90	0.80	7	28.57	0.07%	k=0.8, delta=1, L=30, z_trend=0
L+rho	k_init, delta_init, L, rho	55.09%	77.29%	-12.52%	-18.76%	0.79	0.66	11	29.55	-14.15%	k_init=0.8, delta_init=1, L=30, rho=0.1
a+z_trend	k, delta, a, z_trend	60.07%	77.29%	-9.71%	-18.76%	0.83	0.71	11	25.27	-9.17%	k=1.2, delta=1, a=0.02, z_trend=0
a+rho	k_init, delta_init, a, rho	60.24%	77.29%	-9.62%	-18.76%	0.86	0.71	13	29.15	-9.01%	k_init=0.8, delta_init=3, a=0.02, rho=0.25
z_trend+rho	k_init, delta_init, z_trend, rho	70.67%	77.29%	-3.73%	-18.76%	0.94	0.82	9	25.44	1.42%	k_init=0.8, delta_init=3, z_trend=0, rho=0.1

GUI-Style Continuous Online Results

Experiment	Swept	Real.	MtM	BH	Gap	Max DD	Sharpe	Calmar	Trades	Avg hold	Δ MtM	End live
baseline	fixed GUI defaults	42.05%	51.30%	83.58%	-17.59%	-25.56%	0.60	0.34	8	122.88	0.00%	end k=0.8, end d=2
q+alpha	q, alpha	101.55%	119.38%	83.58%	19.50%	-17.68%	1.07	0.97	16	57.19	68.08%	end k=0.8, end d=2
q+beta	q, beta	86.48%	102.98%	83.58%	10.57%	-17.68%	0.96	0.86	15	65.33	51.68%	end k=0.8, end d=2
q+L	q, L	80.98%	96.99%	83.58%	7.30%	-17.68%	0.92	0.83	14	72.07	45.69%	end k=0.8, end d=2
q+a	q, a	80.98%	96.99%	83.58%	7.30%	-17.68%	0.92	0.83	14	72.07	45.69%	end k=0.8, end d=2
q+z_trend	q, z_trend	70.30%	85.37%	83.58%	0.97%	-18.76%	0.83	0.70	14	77.21	34.07%	end k=0.8, end d=2
q+rho	q, rho	94.84%	112.08%	83.58%	15.52%	-19.50%	1.03	0.84	14	73.64	60.78%	end k=0.8, end d=3
alpha+beta	alpha, beta	49.87%	63.13%	83.58%	-11.14%	-25.56%	0.68	0.40	9	119.22	11.83%	end k=0.8, end d=2
alpha+L	alpha, L	49.87%	63.13%	83.58%	-11.14%	-25.56%	0.68	0.40	9	119.22	11.83%	end k=0.8, end d=2
alpha+a	alpha, a	77.09%	87.27%	83.58%	2.01%	-22.97%	0.87	0.58	20	49.75	35.98%	end k=0.8, end d=2
alpha+z_trend	alpha, z_trend	54.59%	68.27%	83.58%	-8.34%	-24.10%	0.72	0.46	9	119.78	16.97%	end k=0.8, end d=2
alpha+rho	alpha, rho	99.72%	112.73%	83.58%	15.88%	-19.50%	1.07	0.84	15	61.93	61.43%	end k=0.8, end d=3
beta+L	beta, L	49.87%	63.13%	83.58%	-11.14%	-25.56%	0.68	0.40	9	119.22	11.83%	end k=0.8, end d=2
beta+a	beta, a	70.85%	79.85%	83.58%	-2.03%	-22.97%	0.90	0.54	19	49.53	28.55%	end k=0.8, end d=2
beta+z_trend	beta, z_trend	54.59%	68.27%	83.58%	-8.34%	-24.10%	0.72	0.46	9	119.78	16.97%	end k=0.8, end d=2
beta+rho	beta, rho	99.72%	112.57%	83.58%	15.79%	-19.50%	1.07	0.84	15	61.93	61.27%	end k=0.8, end d=3
L+a	L, a	77.09%	87.27%	83.58%	2.01%	-22.97%	0.87	0.58	20	49.75	35.98%	end k=0.8, end d=2
L+z_trend	L, z_trend	74.34%	75.58%	83.58%	-4.36%	-24.64%	0.76	0.49	10	119.30	24.29%	end k=0.8, end d=2
L+rho	L, rho	99.72%	112.73%	83.58%	15.88%	-19.50%	1.07	0.84	15	61.93	61.43%	end k=0.8, end d=3
a+z_trend	a, z_trend	83.46%	94.01%	83.58%	5.68%	-22.97%	0.91	0.62	21	48.71	42.71%	end k=0.8, end d=2
a+rho	a, rho	99.72%	112.73%	83.58%	15.88%	-19.50%	1.07	0.84	15	61.93	61.43%	end k=0.8, end d=3
z_trend+rho	z_trend, rho	102.48%	115.09%	83.58%	17.16%	-19.50%	1.08	0.85	15	62.20	63.79%	end k=0.8, end d=3

High-Level Takeaways

- Stitched best MtM return: z_trend+rho with 70.67%.
- Stitched best benchmark gap: z_trend+rho with -3.73%.
- Stitched best Sharpe ratio: q+rho with 0.95.
- GUI-style best MtM return: q+alpha with 119.38%.
- GUI-style best benchmark gap: q+alpha with 19.50%.
- GUI-style best Sharpe ratio: z_trend+rho with 1.08.

Caveats

The 4D grid is substantially larger than the 3D grid, so overfitting risk is higher. Treat this as a sensitivity and screening report, not final evidence. Pairs involving ρ are not perfectly comparable to static pairs because they use the online adaptation engine inside folds.